

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. (CL) Examination

December 2012

CL101 Fundamentals of Civil Engineering

Date: 11.12.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is hydraulic structure? Describe gravity dam with neat sketch. [05]
(b) Enlist the branches of civil engineering. [02]
- Q - 2 (a) Define hydrology and explain hydrologic cycle with neat sketch. [06]
(b) Explain in detail (Any Two): [04]
Structural engineering, Geotechnical engineering, Transportation engineering.
(c) Give the functions of the following; [04]
Lintel, Damp proof coarse, Door, Column

OR

- Q - 2 (a) Write note on methods of conservation of water. [06]
(b) Enlist the roles of civil engineer. [04]
(c) Explain Circulation & Roominess in detail. [04]
- Q - 3 **Figure-1** shows line diagram of a plan. Draw its wall thickness plan. Use levels and schedule of opening given below the figure. Assume suitable scale. [14]

OR

- Q - 3 **Figure-1** shows line diagram of a plan. Draw elevation from the face, where arrow is given. Use levels and schedule of opening given below the figure. Assume suitable scale. [14]

SECTION – II

- Q - 4 (a) Show component parts of prismatic compass using a neat sketch. [02]
(b) The length of a survey line measured with a chain was found to be 350 m. calculate [05]
the true length of a line if (i) the length was measured with a 30 m chain and was 10
cm too long, and (ii) the length of chain was 30 m in the beginning and 30.10 m at the
end of work.
- Q - 5 (a) Nine perpendicular offsets were taken from a survey line to an irregular boundary line [05]

at an interval of 5 m. Calculate the area by mid-ordinate rule and average ordinate rule.

Distance(m)	0	5	10	15	20	25	30	35
Offsets (m)	1.20	1.27	2.36	0.95	3.75	1.95	1.38	2.10

- (b) (1) Define Any Two: True meridian, Dip of needle and Fore bearing of a survey line. [05]

(2) Convert following whole circle bearings to quadrantal bearings;

(i) $179^{\circ}30'$ (ii) 45° (iii) $269^{\circ}30'$

- (c) Explain plane surveying and geodetic surveying in detail, with neat sketch. [04]

OR

- Q - 5 (a) Explain working of planimeter with neat sketch. [05]

- (b) The following bearings were observed in running a closed traverse ABCDE. [05]

Calculate the interior angles of the traverse and apply necessary checks.

Line	Fore Bearing
AB	107°
BC	22°
CD	$281^{\circ}30'$
DE	189°
EA	$124^{\circ}30'$

- (c) What is scale? Describe different types of scales in detail. [04]

- Q - 6 (a) Define bench mark. Explain any three types of bench marks in details. [07]

- (b) Give classification of surface transportation. Enlist the advantages of roadways over railways. [07]

OR

- Q - 6 (a) The following consecutive readings were taken with a dumpy level: 3.525, 2.630, 2.835, 2.945, 1.915, 3.315, 3.560, 2.125, 1.895, 0.965 and 1.000. [07]

The instrument was shifted after fourth and ninth readings. The first reading was taken on Bench mark of 100.000 m RL. Use Height of Instrument method to calculate the reduced levels of all points and apply usual checks.

- (b) Enlist the advantages and disadvantages of BOT projects for highways. [07]

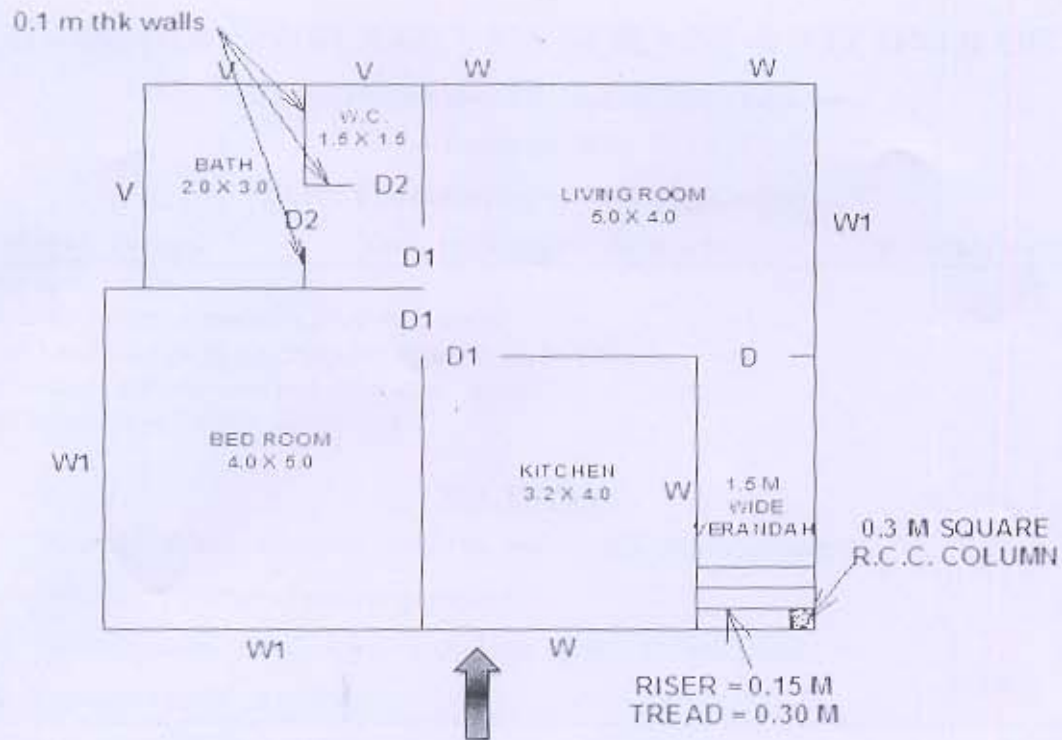


FIGURE - 1

Note: All walls are 30 cm thick; accept where mentioned in above figure.

REQUIRED LEVELS:-

Plinth Level = 0.60 m above Ground Level

Lintel Level = 2.00 m above Plinth Level

Slab Bottom = 3.00 m above Plinth Level

Slab Thickness = 0.10 m thick

Parapet Height = 0.90 m above Slab Top.

SCHEDULE OF OPENING

SR.NO.	DE Scription - SIZE	NO(S)
1.	DOOR 'D' - 1.5 X 2.0	1
2.	DOOR 'D1' - 1.0 X 2.0	3
3.	DOOR 'D2' - 0.9 X 2.0	2
4.	WINDOW 'W' - 1.0 X 1.0	4
5.	WINDOW 'W1' - 1.8 X 1.0	3
6.	VENTILATOR 'V' - 0.6 X 0.6	3